

Data Representation

Lesson Objectives

By the end of this lesson, students should be able to:

- Explain different **number systems** and how data is represented in binary form.
 - Convert numbers between **binary, octal, decimal, and hexadecimal** systems.
 - Describe how **text, images, and sounds** are represented digitally.
 - Write simple **pseudocode** for base conversions.
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1 Introduction to Data Representation

Computers don't understand letters or symbols the way humans do — they understand **numbers**.

All data inside a computer (text, images, audio, video) is represented using **binary digits (bits)**:

0 and **1**

These bits represent the **ON (1)** and **OFF (0)** states of electronic switches inside the hardware.

2 Number Systems

There are several numbering systems used in computing.

Number System	Base	Digits Used	Example
Binary	2	0, 1	101101
Octal	8	0–7	57 ₈
Decimal	10	0–9	123 ₁₀
Hexadecimal	16	0–9, A–F	3A ₁₆

Key Idea:

- Humans use **decimal (base 10)**.

- Computers use **binary (base 2)** because digital circuits have two states: ON/OFF.
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3 Converting Between Number Systems

♦ Binary to Decimal

Multiply each binary digit by 2 raised to its positional power (from right to left).

Example:

Convert 1011_2 to decimal

$$\rightarrow (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$$

$$\rightarrow 8 + 0 + 2 + 1 = \mathbf{11}_{10}$$

♦ Decimal to Binary

Repeatedly divide the decimal number by 2 and record the remainders.

Example:

Convert 13_{10} to binary:

$$13 \div 2 = 6 \text{ remainder } 1$$

$$6 \div 2 = 3 \text{ remainder } 0$$

$$3 \div 2 = 1 \text{ remainder } 1$$

$$1 \div 2 = 0 \text{ remainder } 1$$

Now read remainders **bottom to top** $\rightarrow \mathbf{1101}_2$

♦ Binary to Hexadecimal

Group binary digits into sets of four (from right).

Convert each group to its hexadecimal equivalent.

Binary	Hexadecimal
0000	0
0001	1
0010	2

0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

Example:

$11011101_2 = (1101)(1101) \rightarrow D D \rightarrow DD_{16}$

Character Representation: ASCII and Unicode

Computers use numeric codes to represent characters.

ASCII (American Standard Code for Information Interchange)

- Uses **7 bits** (128 possible characters).
- Represents English letters, digits, and symbols.
- Example:
 - $A \rightarrow 65$
 - $B \rightarrow 66$
 - $a \rightarrow 97$
 - $\text{space} \rightarrow 32$

Unicode

- Uses **16 bits or more**.
- Represents characters from **multiple languages and symbols** worldwide.
- Example:
 - $A \rightarrow U+0041$
 - $\Omega \rightarrow U+03A9$

- 😊 → U+1F60A

5 Image Representation

Images are made up of **pixels** (picture elements).

Each pixel stores color information as a combination of:

- **Red**
- **Green**
- **Blue (RGB)** values.

Each color value is stored in **binary** form.

For example, a black and white image can store pixels as:

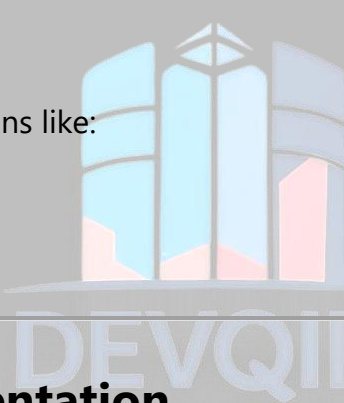
- **0 → black, 1 → white.**

Color images use combinations like:

RGB(255, 0, 0) → Red

RGB(0, 255, 0) → Green

RGB(0, 0, 255) → Blue



6 Sound Representation

Sound waves are **analog**, so they must be converted into **digital form** using a process called **sampling**.

- **Sampling:** Taking sound measurements at regular intervals (samples per second).
- **Sample Rate:** The number of samples per second (e.g., 44,100 Hz for CDs).
- **Bit Depth:** The number of bits used to represent each sample (e.g., 16-bit audio).

The higher the **sample rate** and **bit depth**, the better the sound quality — but the larger the file size.

7 Class Activity: Conversion Practice

Objective:

To apply number system conversion techniques.

Instructions:

1. Convert the following:
 - a. $11010_2 \rightarrow ?_{10}$
 - b. $57_8 \rightarrow ?_2$
 - c. $45_{10} \rightarrow ?_{16}$
 - d. $9C_{16} \rightarrow ?_2$
2. Write a short pseudocode that converts a decimal number to binary.

Example Pseudocode: Decimal to Binary

Algorithm DecimalToBinary
Input: N (a positive integer)
Output: Binary equivalent of N

```
WHILE N > 0
  R ← N MOD 2
  PRINT R
  N ← N DIV 2
END WHILE
REVERSE printed digits
END Algorithm
```



Lab Exercise (Documentation Required)

Task:

Develop pseudocode for converting:

1. Binary $\rightarrow$ Decimal
2. Decimal $\rightarrow$ Hexadecimal
3. Hexadecimal $\rightarrow$ Binary

Documentation:

Your lab report should include:

- A short explanation of each number system
- Your pseudocode for each conversion
- Sample inputs and outputs
- Reflections on what was challenging or interesting

 **Deliverable:** Submit your pseudocode and explanations in a Markdown or Word document.

Git Task

- Update your GitHub learning repository.
- Create a new file: week3_data_representation.md.
- Document:
 - Key notes from this lesson.
 - Your pseudocode for conversions.
 - Screenshots or examples from your lab.

Push your work to GitHub using:

```
git add week3_data_representation.md
git commit -m "Week 3: Data Representation notes and pseudocode"
git push
```